Regression & 1D CNN Result Comparison with AEC, SMI

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1. What's discussed in the last meetings

- Clinic Data： Age, Sex → Age, Sex, BMI
  - Output을 TAMA가 아닌 SMI 사용 (SMI = TAMA/Height^2)
- AEC Features Extraction : 수동 → 자동 파이프라인 (4단계)
  - 단계가 불필요하게 세분화되어 있음

2. Research Summary

- **SMI 데이터 확립 → TAMA, BMI, Height, Weight 등의 데이터로 확인 가능**

- AEC data Resampling → 256 point로 정규화
  - 이후 주요 feature 추출 (mean, std, range, skewness 등 65개 feature)

- **이전 연구와 동일한 케이스로 비교 분석**

Case	Feature 구성	Feature_num
clinic_only	PatientAge, PatientSex, BMI	3
aec_only	aec1 ~ aec256 (원시 AEC 시계열 통계)	256
aec_feature_only	aec_feature_filtered 시트 전체 사용	65
aec_feature_selected	AEC 피처 fold별 선택 (|r|<0.8, VIF<10)	가변
clinic_aec_feature	Age/Sex/BMI (고정) + AEC 피처 fold별 선택	가변

연구 프로토타입 정의

Research Overview – Main Study

- **연구 목적**
  - Opportunistic CT Biomarker 기반 → 5년 내 vertebral compression fracture 예측
  - AEC의 incremental value 검증
- **데이터**
  - Baseline APCT + follow-up imaging 보유 환자
  - 주요 변수: Age, Sex, AEC, Muscle, Bone Density 등
- **모델 구조**
  - Model A: Age + Sex
  - Model B: + Bone
  - Model C: + Muscle
  - Model D: + AEC (핵심 비교)

Research Overview – Pilot Feasibility Study (현재 진행 연구)

- **연구 목적**
  - AEC가 단순 보조 변수인지 vs 독립적인 biomarker인지 검증

  → 본 연구 모델 포함 가능성 평가

- **핵심 질문**
  - AEC는 age/sex 이상 정보를 가지는가?
  - Muscle/Bone과 독립적 연관성이 존재하는가?
  - Scanner/Center 보정 후에도 유지되는가?

Research Overview – Pilot Feasibility Study (현재 진행 연구)

- **분석 전략**
  - 분포 분석 (AEC by 센터, 성별, 연령) – Complete
  - **회귀 분석 (muscle area / attenuation / bone density ~ AEC + age + sex)**
  - **고급 분석**
    - **Multivariable (center, scanner 보정)**
    - Spline 분석 → 비선형 관계 확인
    - Center-stratified 분석

- **결과물**
  - AEC distribution plot
  - AEC vs biomarker scatter
  - Regression Table
  - Stratified Result Figure 등

데이터 정리

3. Detailed progress

- **Data – 강남**
  - Clinic Data 수 : 3,672 → 중복 PatientID 제거 : 2,025 → TAMA outlier(0, 1), BMI 제거 : 1,642
  - Axial Data 수 : 1,642 → Fixed Tube Current Value 제거 : 1,373

총 환자 수

1,373명

남성

514명 (37.4%)

여성

859명 (62.6%)

SMI 분포

성별	N	mean	std	P25	median
Female	859	42.33	6.23	38.46	41.64
Male	514	51.75	8.71	46.66	52.13

TAMA 분포

성별	N	mean	std	P25	median
Female	859	104.94	15.31	95	103
Male	514	148.23	26.35	132	148

3. Detailed progress

- 성별에 따른 TAMA, SMI 하위 25% 데이터 분포

- TAMA 하위 25%인 데이터 수 : 358 / 1,373
  - M : 136 / 514, F ; 222 / 859

- SMI 하위 25%인 데이터 수 : 344 / 1,373
  - M : 129 / 514, F : 215 / 859

- TAMA, SMI 모두 하위 25%인 데이터 수 : 259 / 1,373
  - M : 101 / 514, F : 158 / 859

3. Detailed progress

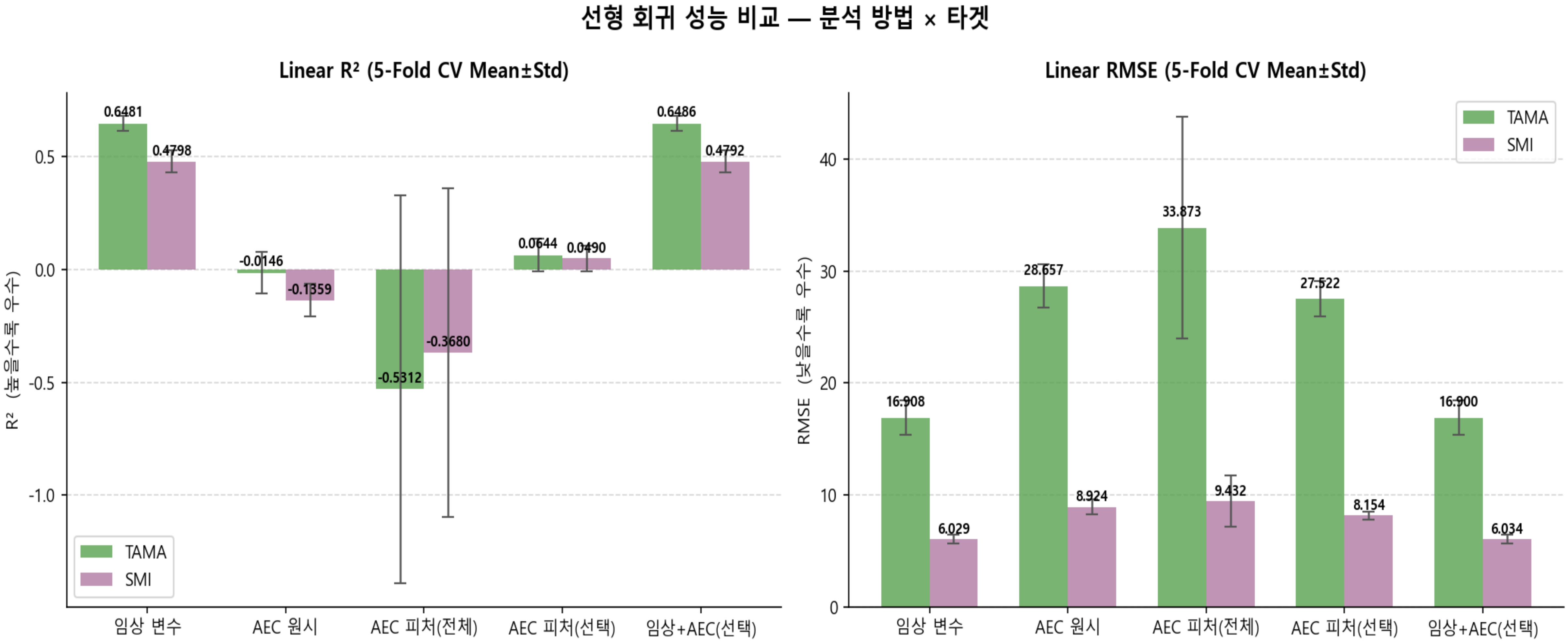
- **데이터 분포 확인**

구분	Mean	Std	Min	p25	Median	p75	max
n slices	168.6	32.0	44	151	165	184	460
z range (mm)	469.6	52.6	215.0	441.0	467.5	497.5	918.0

구분	성별	Mean	Std	Min	Median	Max
n slices (개수)	F	168.61	32.05	50.0	165.0	460.0
	M	168.65	32.04	44.0	166.0	405.0
z range (mm) (범위)	F	470.97	51.47	234.0	468.0	918.0
	M	467.58	54.34	215.0	465.0	808.0

3. Detailed progress

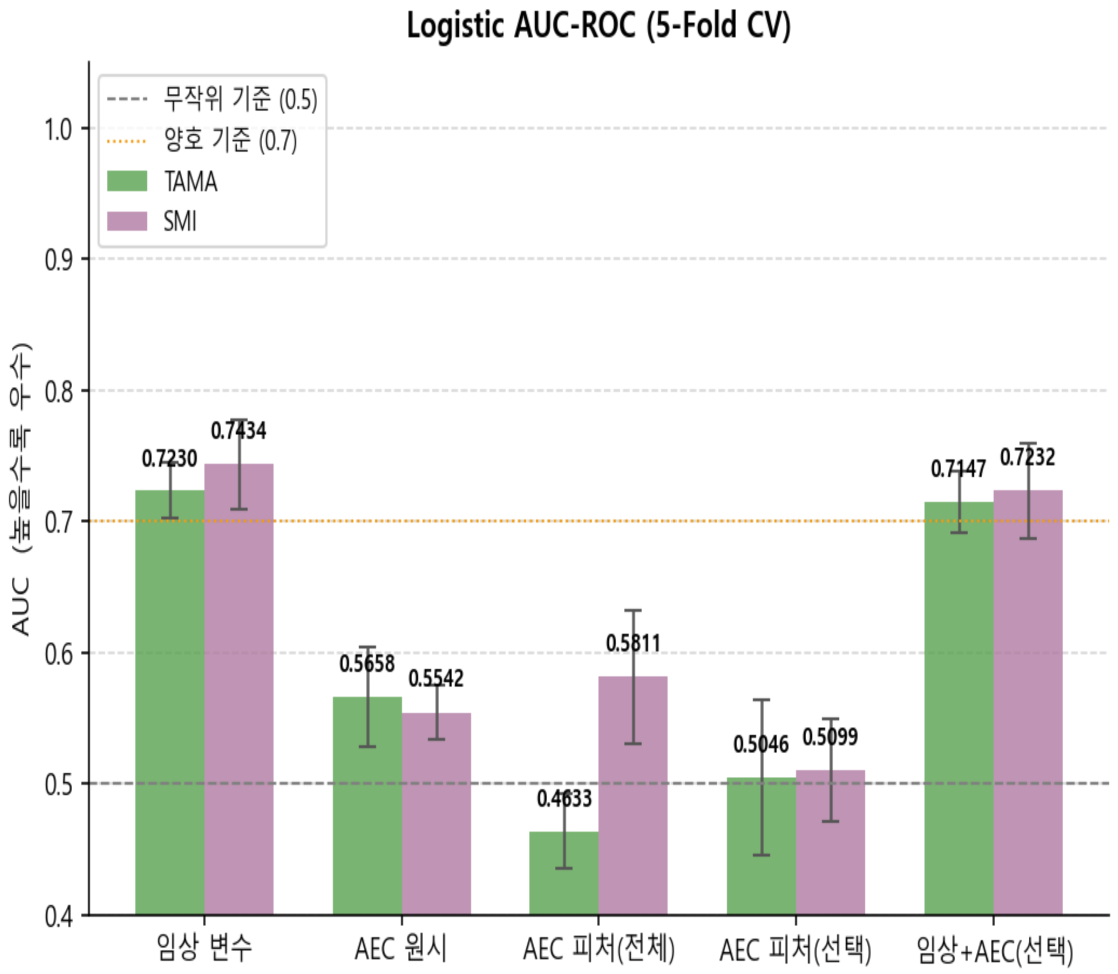
- Case 별 성능 비교

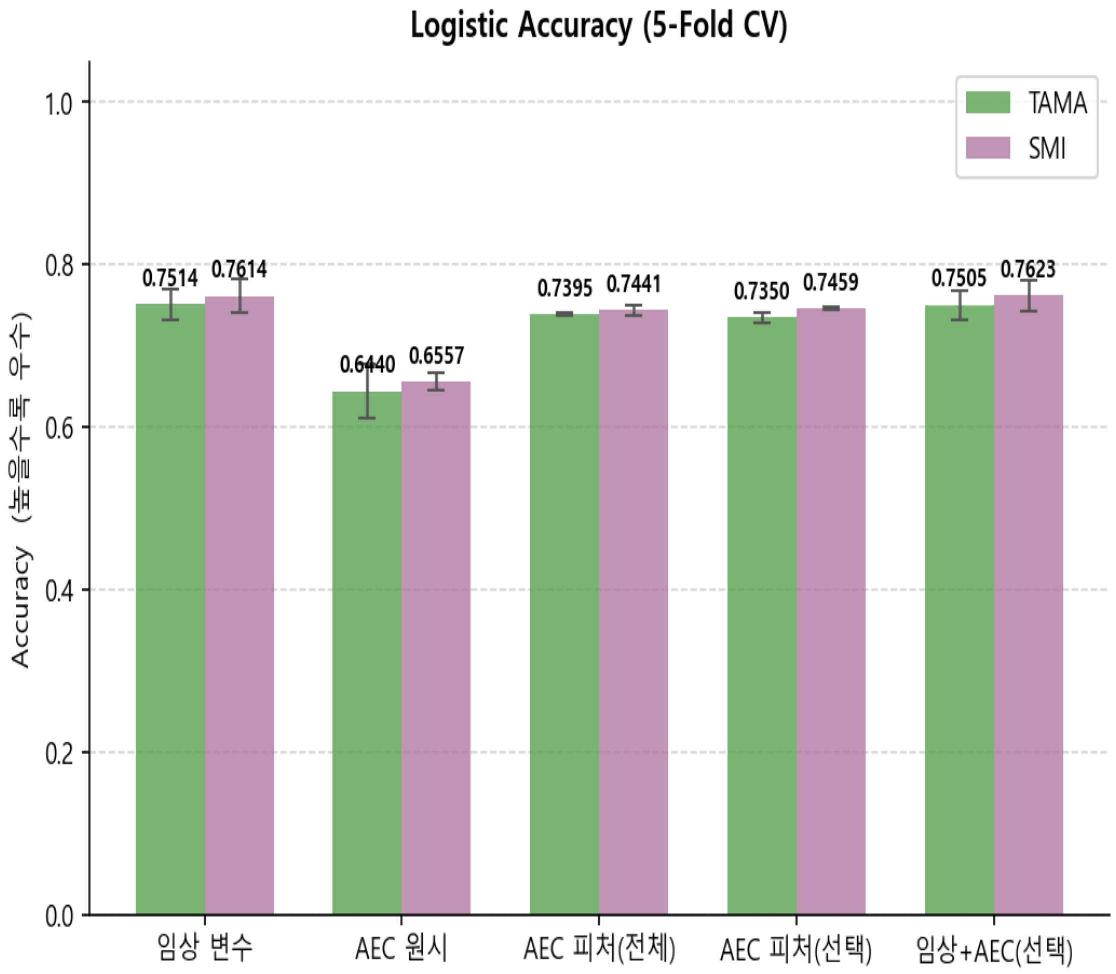


3. Detailed progress

- Case 별 성능 비교

로지스틱 회귀 성능 비교 — 분석 방법 × 타겟



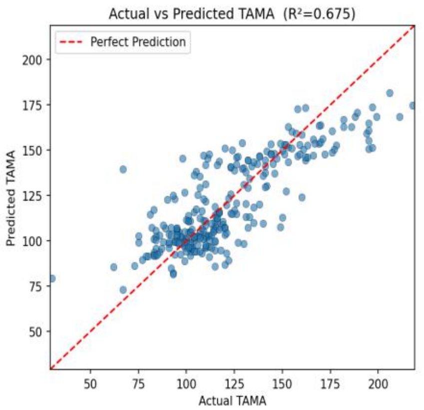


3. Detailed progress

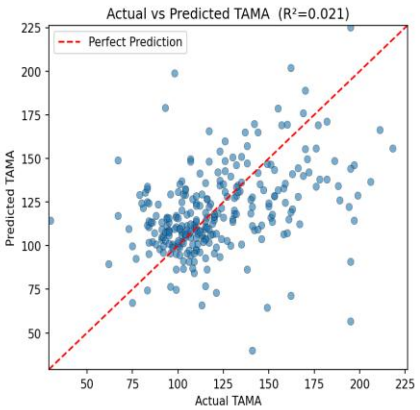
- **Case 별 성능 비교 – Linear Regression (TAMA)**

방법	CV R^2 (mean ± std)	CV RMSE (mean ± std)	Test R^2	Test RMSE	피처
임상 변수(Age, Sex, BMI)	**0.6481 ± 0.0331**	**16.908 ± 1.528**	**0.6750**	**17.629**	PatientSex, PatientAge, BMI (3개)
AEC raw data 256개	-0.0146 ± 0.0910	28.657 ± 1.929	0.0212	30.594	aec1 ~ aec256 (256개)
AEC feature 전체	-0.5312 ± 0.8611	33.873 ± 9.896	0.2098	27.488	signal_length ~ wavelet_energy_ratio_D1 (65개)
AEC feature 선택	0.0644 ± 0.0724	27.522 ± 1.582	0.0617	29.955	AEC 선택 피처 (|r|<0.8, VIF<10.0) — Train 기준 14개
임상 변수 + AEC feature 선택	**0.6486 ± 0.0316**	**16.900 ± 1.535**	**0.6765**	**17.587**	PatientSex, PatientAge, BMI (고정) + AEC 선택 피처

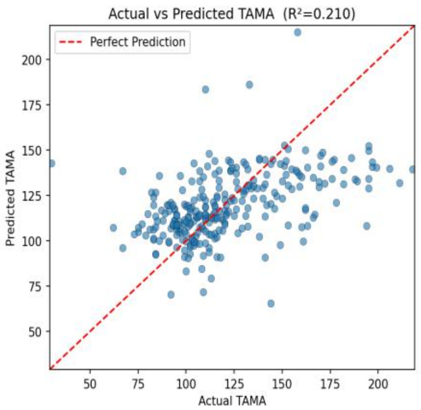
임상 변수 (Age, Sex, BMI)



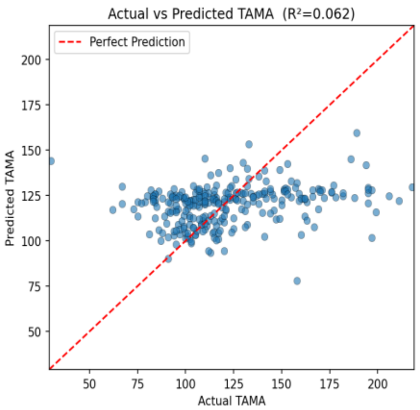
AEC raw data 256개



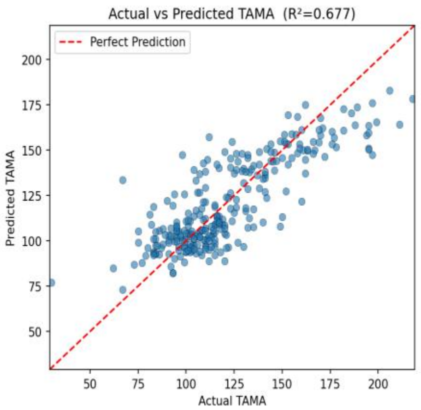
AEC feature 전체



AEC feature 선택



임상 변수 + AEC feature 선택



3. Detailed progress

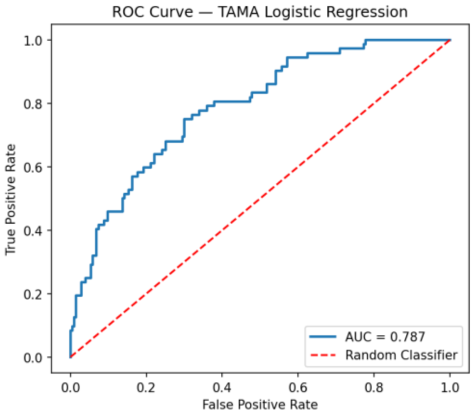
- **Case 별 성능 비교 – Logistic Regression (TAMA)**

방법	CV AUC (mean ± std)	CV Acc (mean ± std)	Test AUC	Test Acc
임상 변수(Age, Sex, BMI)	**0.7230 ± 0.0212**	**0.7514 ± 0.0192**	**0.7869**	**0.7636**
AEC raw data 256개	0.5658 ± 0.0378	0.6440 ± 0.0336	0.5855	0.6327
AEC feature 전체	0.4633 ± 0.0285	0.7395 ± 0.0017	0.5099	0.7382
AEC feature 선택	0.5046 ± 0.0593	0.7350 ± 0.0066	0.6388	0.7382
임상 변수 + AEC feature 선택	**0.7147 ± 0.0239**	**0.7505 ± 0.0184**	**0.7712**	**0.7782**

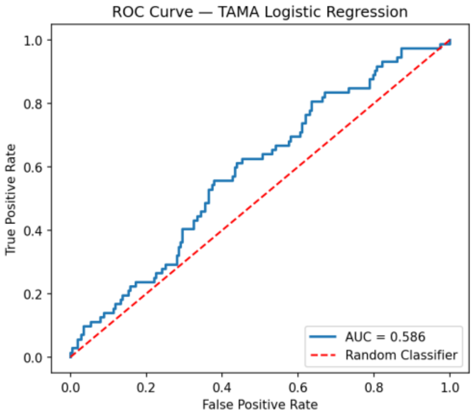
3. Detailed progress

- Case 별 성능 비교 – Logistic Regression (TAMA)

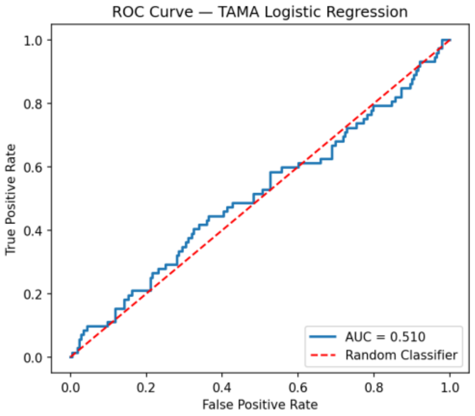
임상 변수 (Age, Sex, BMI)



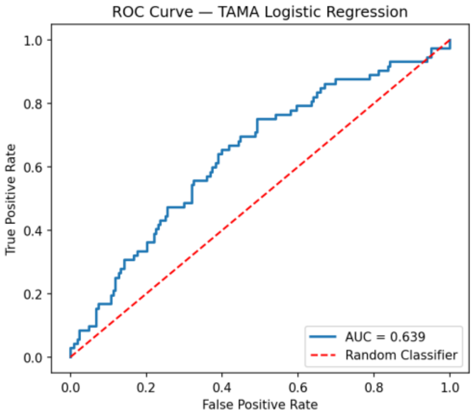
AEC raw data 256개



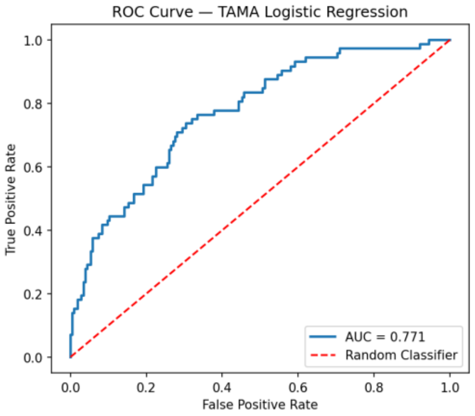
AEC feature 전체



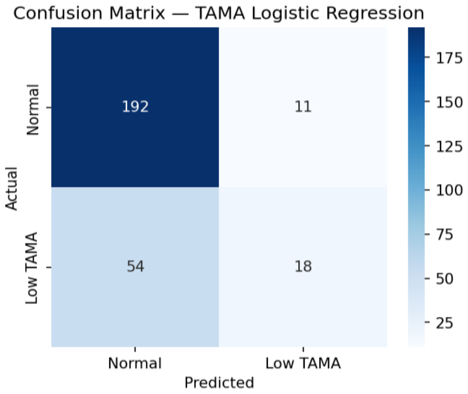
AEC feature 선택



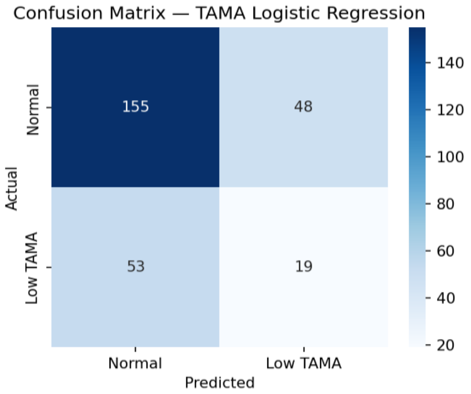
임상 변수 + AEC feature 선택



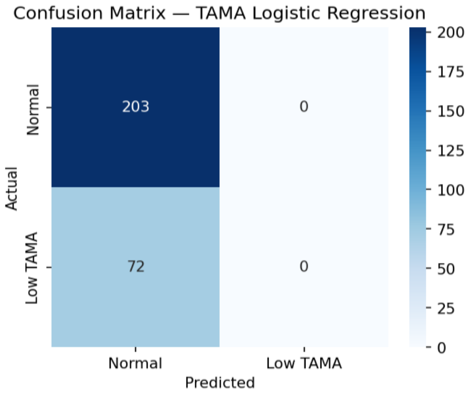
임상 변수 (Age, Sex, BMI)



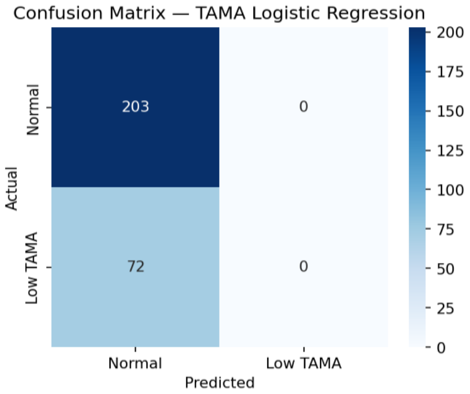
AEC raw data 256개



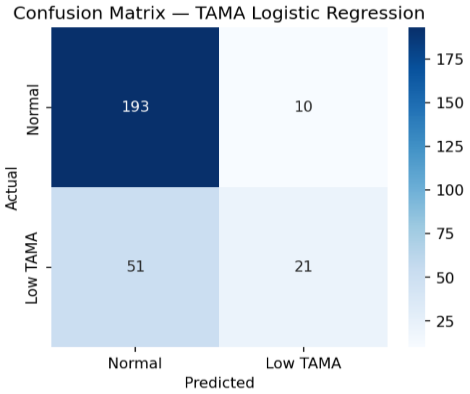
AEC feature 전체



AEC feature 선택



임상 변수 + AEC feature 선택

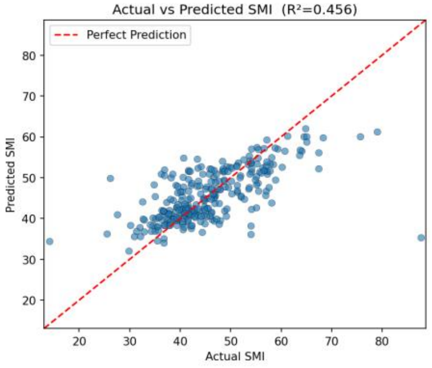


3. Detailed progress

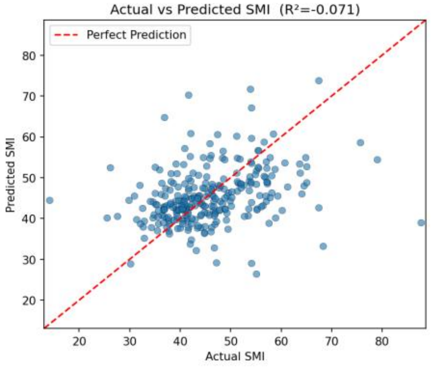
- **Case 별 성능 비교 – Linear Regression (SMI)**

방법	CV R² (mean ± std)	CV RMSE (mean ± std)	Test R²	Test RMSE	피처
임상 변수(Age, Sex, BMI)	**0.4798 ± 0.0494**	**6.029 ± 0.404**	**0.4564**	**6.785**	PatientSex, PatientAge, BMI (3개)
AEC raw data 256개	-0.1359 ± 0.0736	8.924 ± 0.647	-0.0709	9.523	aec1 ~ aec256 (256개)
AEC feature 전체	-0.3680 ± 0.7289	9.432 ± 2.286	0.1555	8.457	signal_length ~ wavelet_energy_ratio_D1 (65개)
AEC feature 선택	0.0490 ± 0.0552	8.154 ± 0.338	0.0186	9.116	AEC 선택 피처 (|r|<0.8, VIF<10.0) — Train 기준 14개
임상 변수 + AEC feature 선택	**0.4792 ± 0.0484**	**6.034 ± 0.408**	**0.4573**	**6.779**	PatientSex, PatientAge, BMI (고정) + AEC 선택 피처

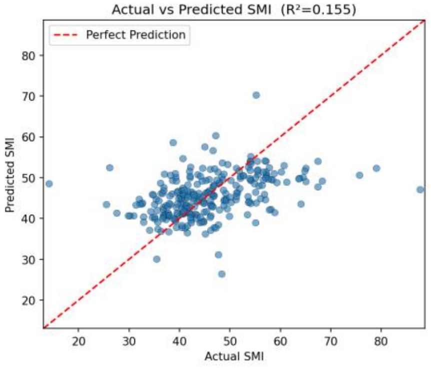
임상 변수 (Age, Sex, BMI)



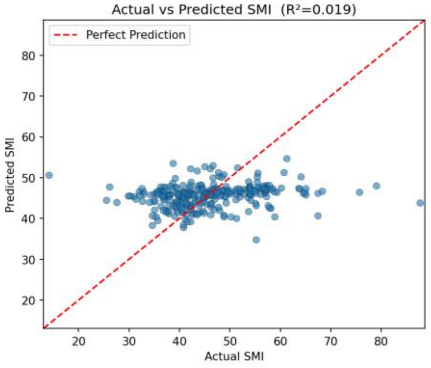
AEC raw data 256개



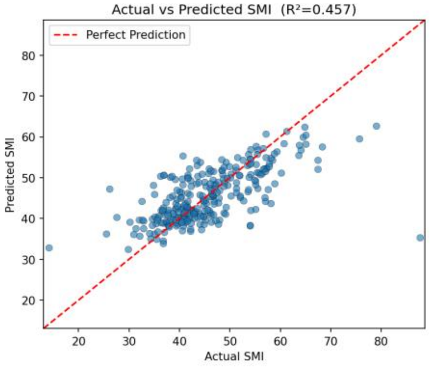
AEC feature 전체



AEC feature 선택



임상 변수 + AEC feature 선택



3. Detailed progress

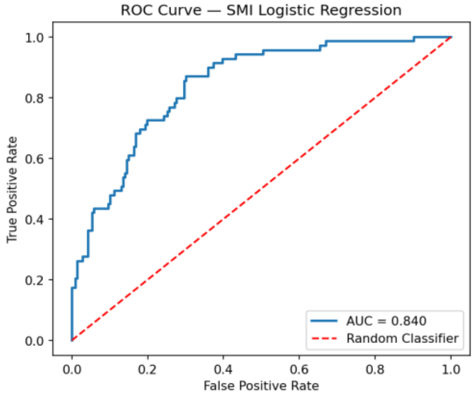
- **Case 별 성능 비교 – Logistic Regression (SMI)**

방법	CV AUC (mean ± std)	CV Acc (mean ± std)	Test AUC	Test Acc
임상 변수(Age, Sex, BMI)	**0.7434 ± 0.0339**	**0.7614 ± 0.0202**	**0.8400**	**0.7964**
AEC raw data 256개	0.5542 ± 0.0209	0.6557 ± 0.0109	0.5651	0.6509
AEC feature 전체	0.5811 ± 0.0504	0.7441 ± 0.0065	0.6287	0.7527
AEC feature 선택	0.5099 ± 0.0392	0.7459 ± 0.0015	0.5323	0.7491
임상 변수 + AEC feature 선택	**0.7232 ± 0.0364**	**0.7623 ± 0.0191**	**0.8326**	**0.7927**

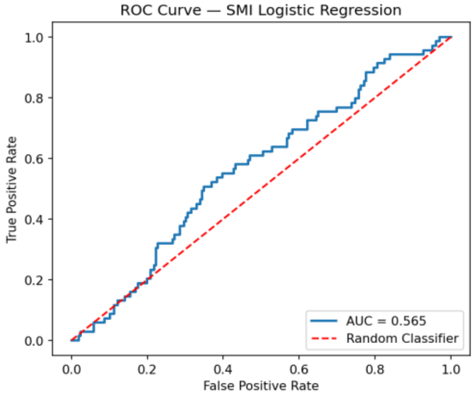
3. Detailed progress

- Case 별 성능 비교 – Logistic Regression (SMI)

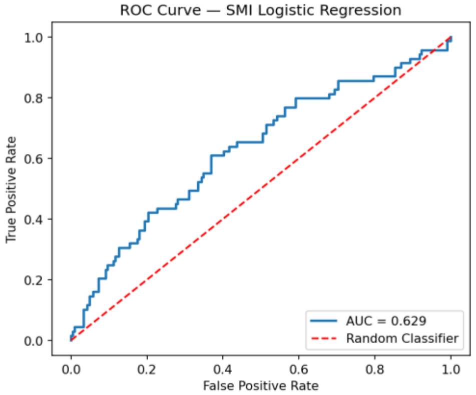
임상 변수 (Age, Sex, BMI)



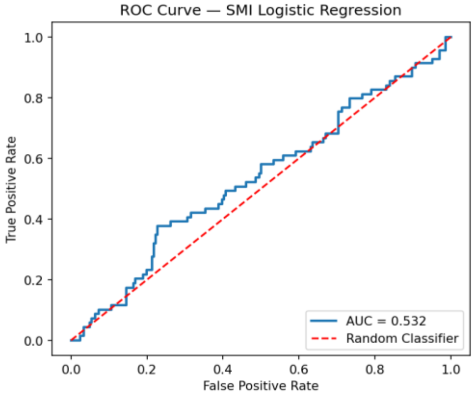
AEC raw data 256개



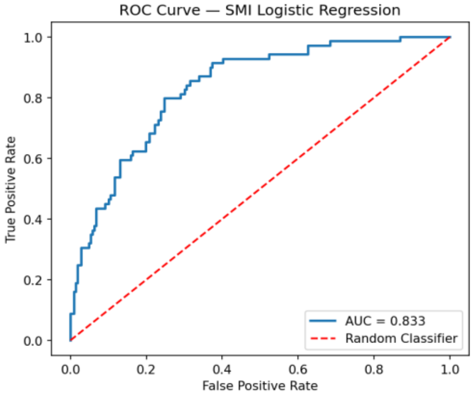
AEC feature 전체



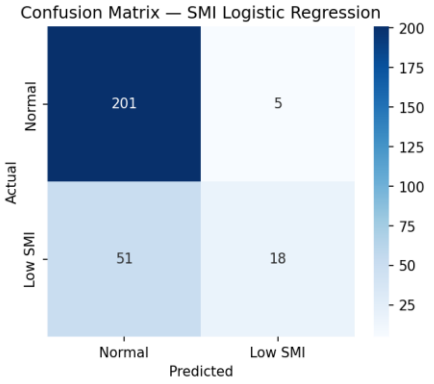
AEC feature 선택



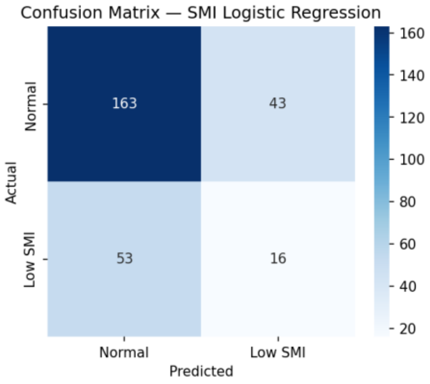
임상 변수 + AEC feature 선택



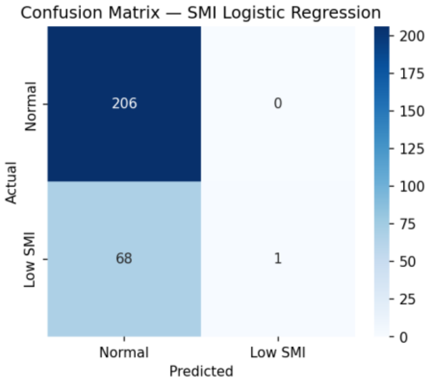
임상 변수 (Age, Sex, BMI)



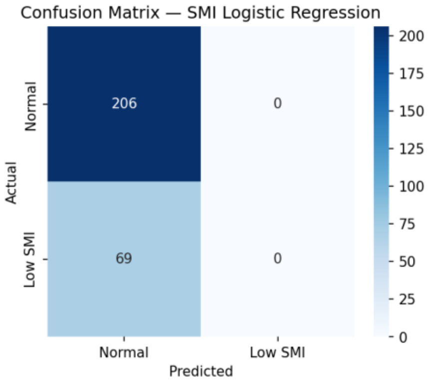
AEC raw data 256개



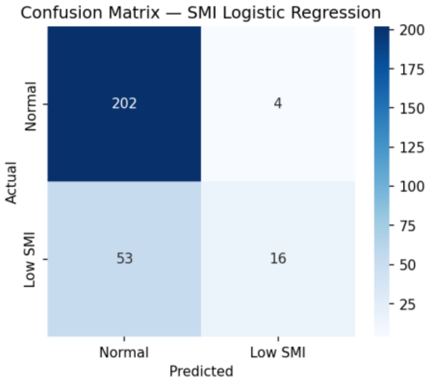
AEC feature 전체



AEC feature 선택



임상 변수 + AEC feature 선택



3. Detailed progress

- **1D CNN(ResNet1D) Code Flow**

1. feature_selector(df_fold_tr) → if 피처 선택 (fold의 학습셋에만 의존)

2. X shape 변환: (N, F) → (N, 1, F) [1D Conv 입력 형식]

3. ResNet1D 초기화 → _fit() 호출

4. best_state 복원 → _eval_epoch (검증셋)

5. MAE, R^2, 성별별 ACC/AUC 계산 및 저장

3. Detailed progress

- **Case 별 성능 비교 – ResNet1D (TAMA)**

방법	CV MAE (mean ± std)	CV R^2 (mean ± std)	Test MAE	Test R^2
임상 변수(Age, Sex, BMI)	12.3331 ± 0.8795	0.6734 ± 0.0422	12.8148	0.7198
AEC raw data 256개	16.9969 ± 1.1511	0.3803 ± 0.0470	17.6174	0.3910
AEC feature 선택	18.4973 ± 1.0157	0.2533 ± 0.0479	19.5603	0.2547
임상 변수 + AEC feature 선택	13.4161 ± 0.4913	0.6069 ± 0.0409	13.8909	0.6662

3. Detailed progress

- Case 별 성능 비교 – ResNet1D (TAMA)

방법	CV ACC (mean ± std)	CV AUC (mean ± std)	Test ACC	Test AUC
임상 변수(Age, Sex, BMI)	0.7604 ± 0.0250	0.6297 ± 0.0529	0.7636	0.6374
AEC raw data 256개	0.7067 ± 0.0186	0.6098 ± 0.0474	0.6945	0.5766
AEC feature 선택	0.6994 ± 0.0347	0.6129 ± 0.0351	0.6909	0.5751
임상 변수 + AEC feature 선택	0.7559 ± 0.0343	0.6060 ± 0.0255	0.7709	0.6293

3. Detailed progress

- Case 별 성능 비교 – ResNet1D (SMI)

방법	CV MAE (mean ± std)	CV R^2 (mean ± std)	Test MAE	Test R^2
임상 변수(Age, Sex, BMI)	4.3631 ± 0.1774	0.5150 ± 0.0402	4.6291	0.4953
AEC raw data 256개	5.3853 ± 0.2697	0.2936 ± 0.0450	5.7357	0.2965
AEC feature 선택	6.0599 ± 0.1140	0.1331 ± 0.0225	6.1585	0.1605
임상 변수 + AEC feature 선택	4.9570 ± 0.2059	0.3705 ± 0.0611	5.6656	0.2995

3. Detailed progress

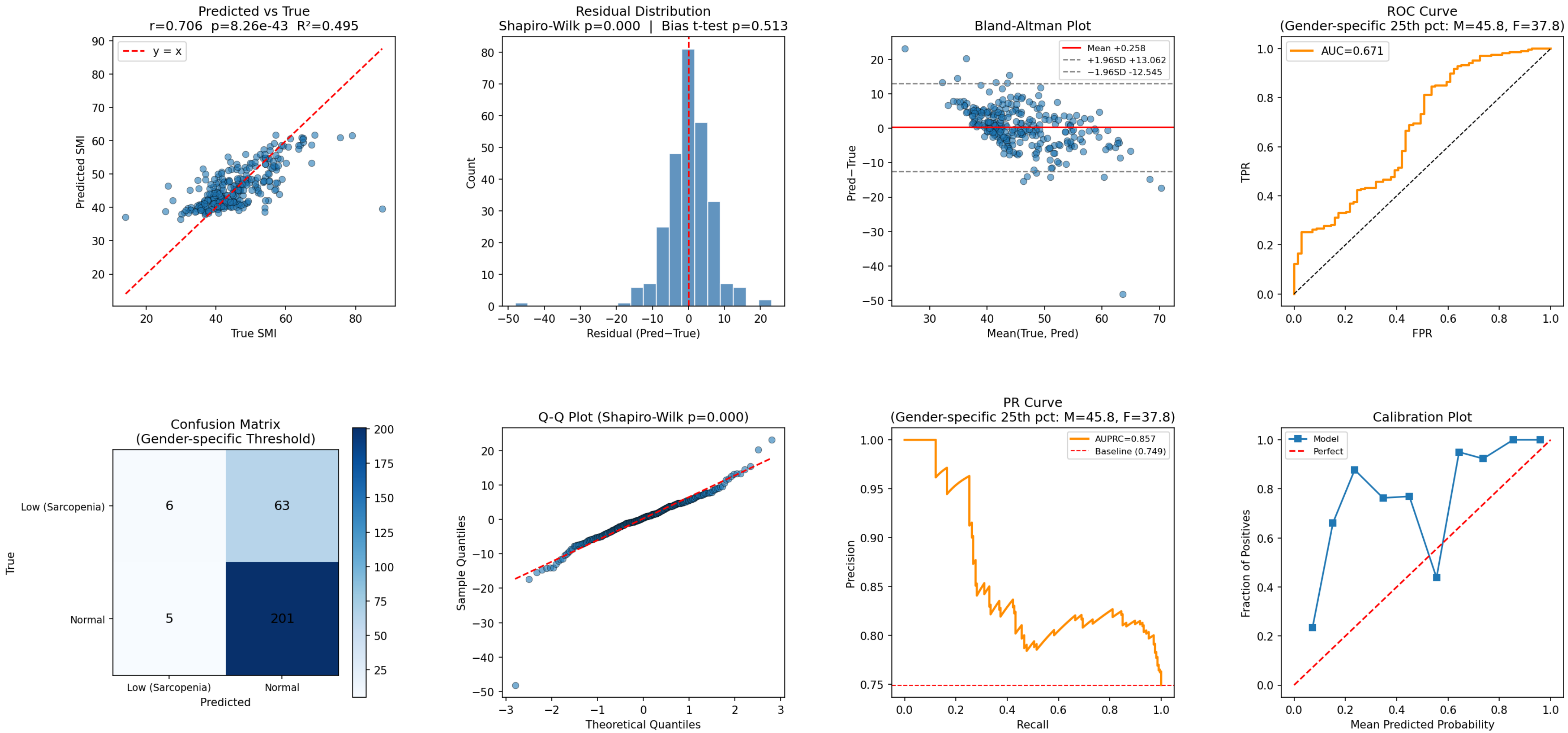
- **Case 별 성능 비교 – ResNet1D (SMI)**

방법	CV ACC (mean ± std)	CV AUC (mean ± std)	Test ACC	Test AUC
임상 변수(Age, Sex, BMI)	0.7659 ± 0.0150	0.6900 ± 0.0370	0.7527	0.6715
AEC raw data 256개	0.7241 ± 0.0249	0.6563 ± 0.0342	0.7345	0.6107
AEC feature 선택	0.6813 ± 0.0368	0.5861 ± 0.0440	0.6873	0.5574
임상 변수 + AEC feature 선택	0.7368 ± 0.0181	0.6837 ± 0.0546	0.7382	0.7283

3. Detailed progress

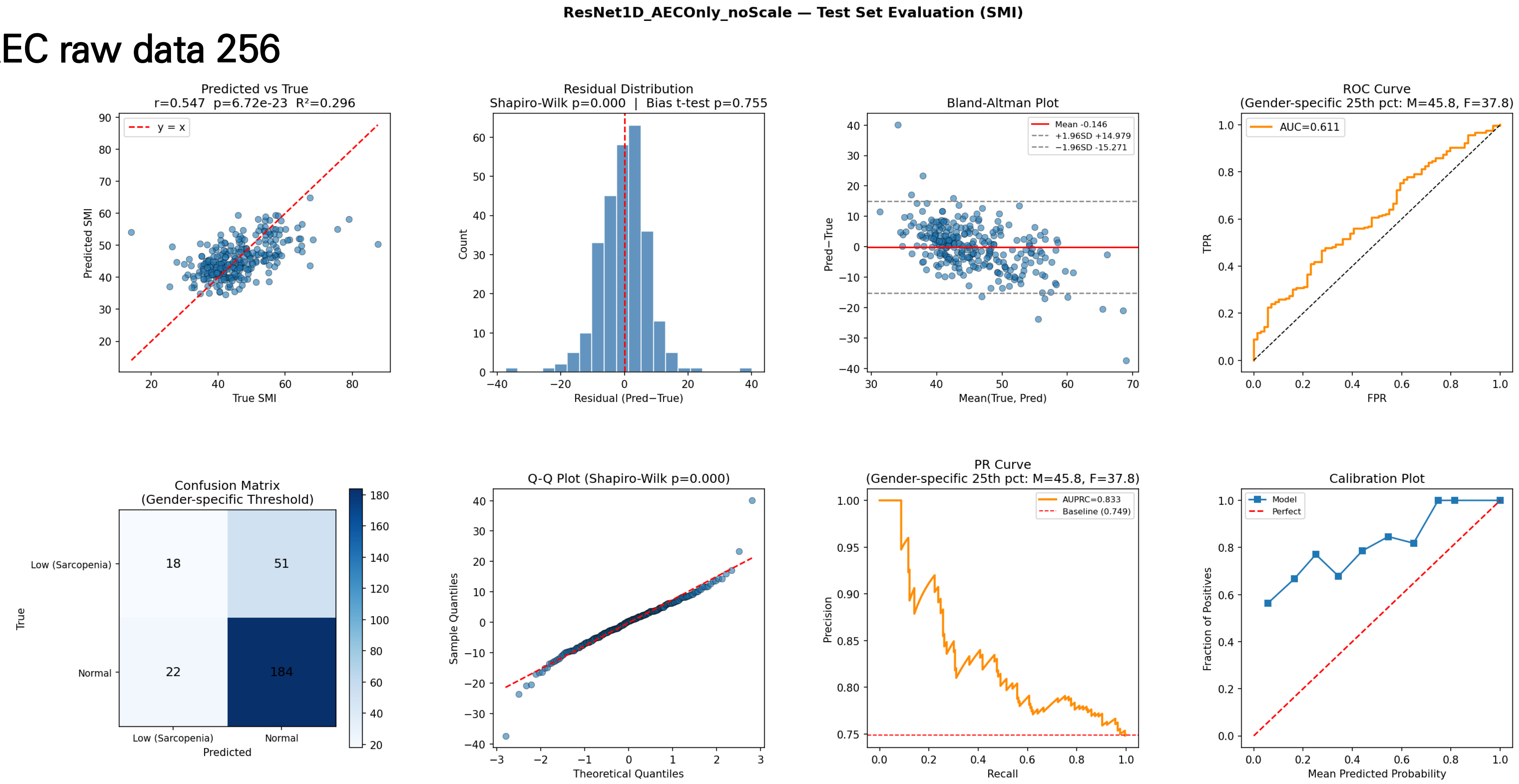
ResNet1D_ClinicOnly_noScale — Test Set Evaluation (SMI)

- Only Clinic



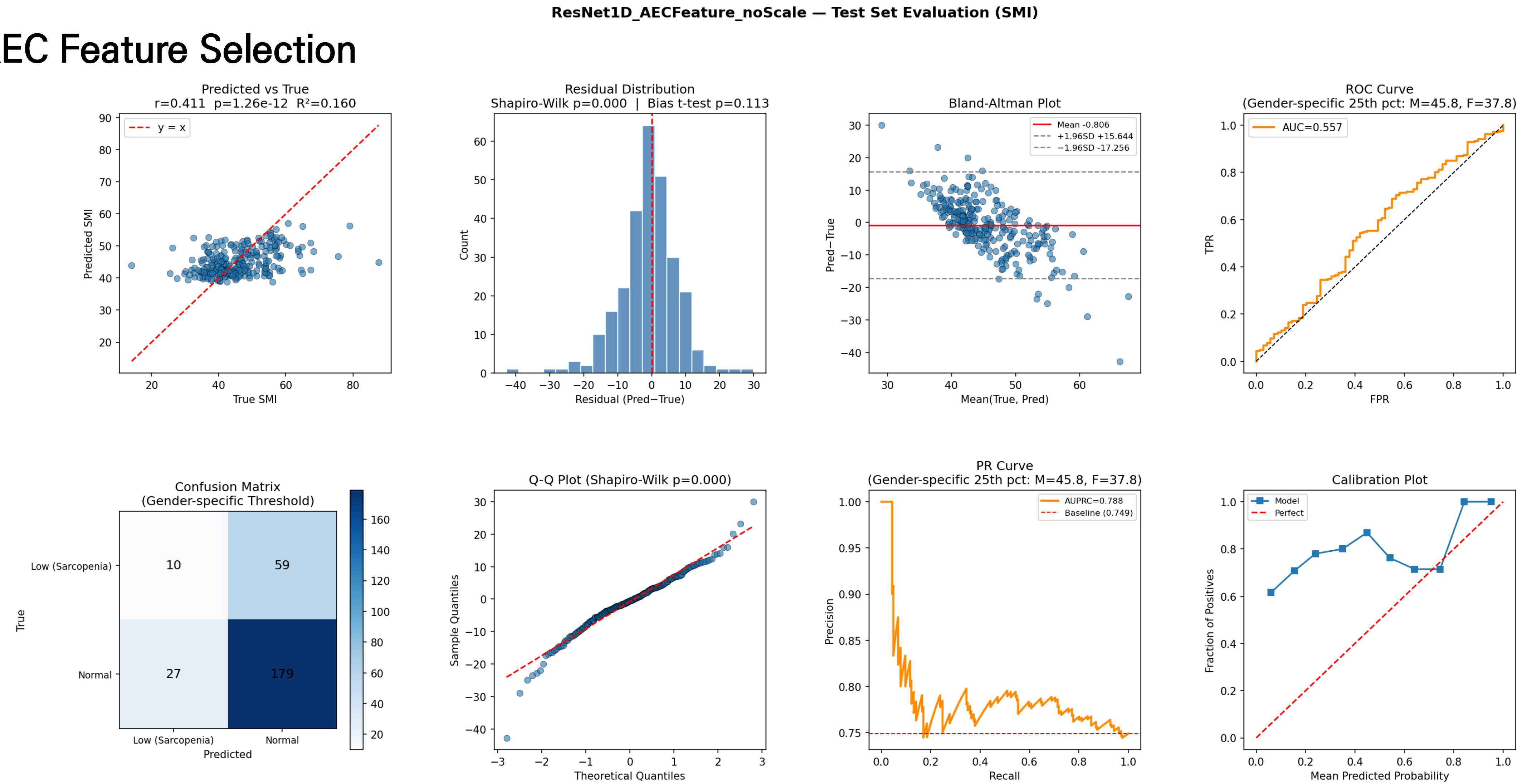
3. Detailed progress

- AEC raw data 256



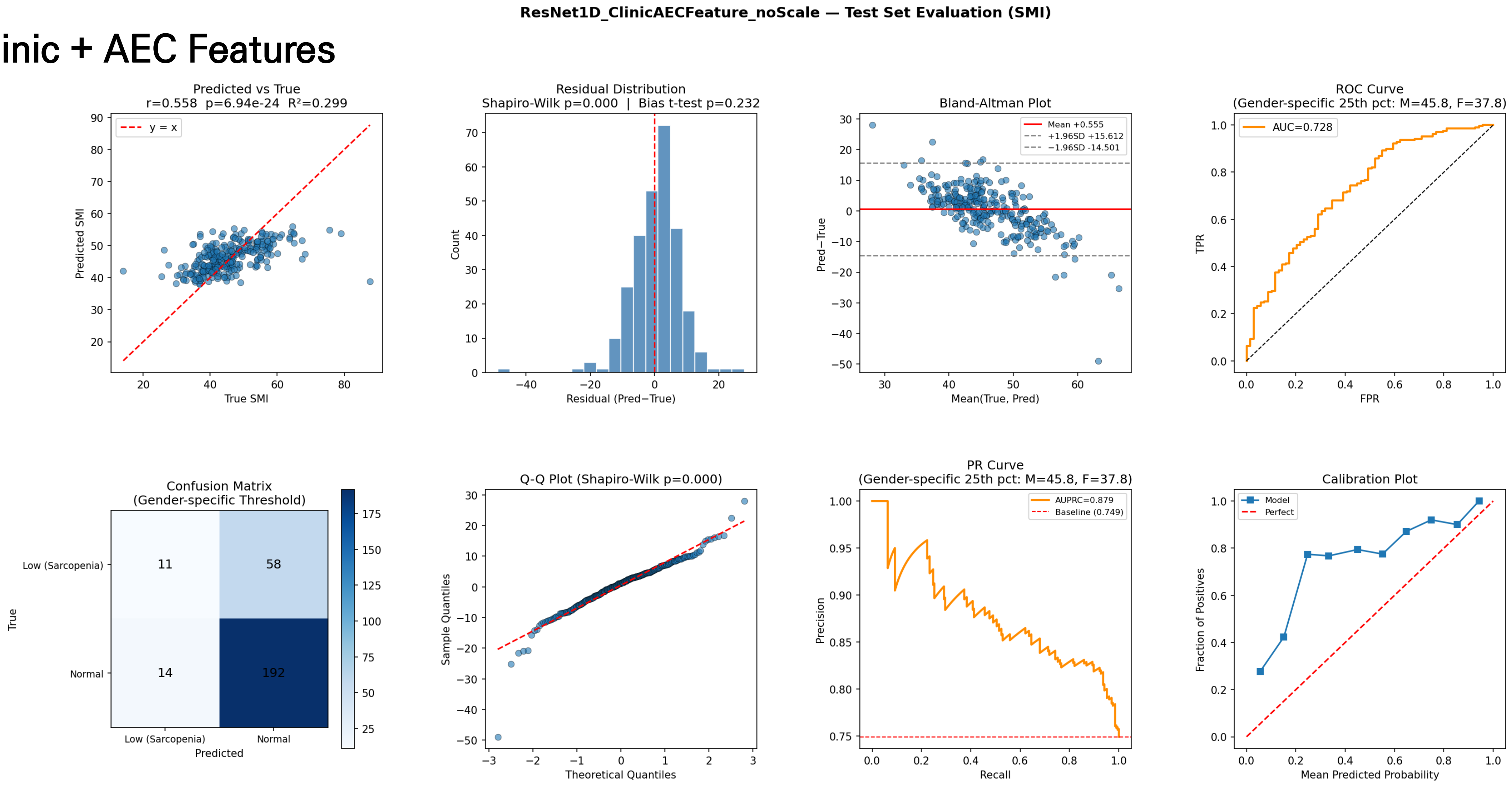
3. Detailed progress

- AEC Feature Selection



3. Detailed progress

- Clinic + AEC Features



4. Conclusion

- **Linear & Logistic Regression 성능 비교 결과는 AEC Feature의 유무에 따른 성능 차이 없음.**
  - Test Dataset AUC 기준 0.8400 → 0.8326 로 ⊿AUC = −0.0074
- **1D CNN에서의 성능 비교 결과는 AEC Feature의 유무에 따른 성능 차이 나타남.**
  - Test Dataset AUC 기준 0.671 → 0.728 로 ⊿AUC = +0.057

5. Next Plan

- **1D CNN에 대한 알고리즘 최적화 진행**
- **Sensitivity analysis 다음 분석 추가진행**
    - full curve 0–100%, central 80% crop, central 60% crop
    - raw AEC curve, within-curve normalized AEC curve
    - scan length extreme case 제외
- **강남 데이터에서 성능확인 후 외부(신촌) 데이터로 validation 진행**

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